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IN-VITRO ENZYME INHIBITION APPROACH FOR PHARMACOLOGICAL MANAGEMENT OF TYPE-2 DIABETES MELLITUS USING A COMBINATION OF QUERCETIN AND GALLIC ACID

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ABSTRACT:

This study was conveyed out to estimate the in vitro inhibitory activity of Quercetin, Gallic acid, Acarbose, and their combination against α-amylase and α-glucosidase enzymes utilizing spectrophotometric methods for pharmacological management of Type-2 diabetes mellitus. In the α-amylase assay, a concentration-dependent rise in percentage inhibition was obeyed for Acarbose and Quercetin. Acarbose exhibited the loftiest inhibitory activity with percentage inhibition categorizing from approximately 60% to 80%, postdated by Quercetin displaying inhibition between 54% and 72%. The combination of Quercetin and Gallic acid laid out temperate activity with inhibition placing from 44% to 58%. However, Gallic acid simply did not show a coherent concentration-dependent trend and displayed abnormal inhibition values. The IC_{50} value for the combination was set up to be roughly 110–120 µg/mL, while Acarbose and Quercetin displayed IC_{50} values in the range of approximately 60–70 µg/mL and 90–100 µg/mL, independently. In the α-glucosidase assay, the effects did not show a harmonious concentration-dependent figure. The control displayed extraordinary absorbance behavior, and the inhibition valuations for test samples were flexible and unreliable. Thus, the α-glucosidase sequels could not be demonstrated conclusively. In conclusion, the study established that Quercetin and Acarbose possessed meaningful α-amylase inhibitory activity, whereas the combination flaunted moderate effects. The α-glucosidase assay results were incompatible and needed further investigation under optimized experimental conditions.

Keywords: *α-amylase, α-glucosidase, Quercetin, Gallic acid, Acarbose, IC_{50}, Enzyme inhibition.*

Introduction

Diabetes mellitus is a habitual metabolic disorder charactered by patient hyperglycemia performing from defects in insulin secretion, insulin action, or both. It is accounted one of the most conventional non-communicable diseases worldwide and has surfaced as a considerable public health interest. Diabetes is astronomically categorized into Type 1 diabetes, Type 2 diabetes, and gestational diabetes. The autoimmune destruction of the pancreatic β-cells results in the complete absence of insulin from the body. In contrast, 90-95% of the population suffering from Type 2 diabetes suffers from resistance to insulin and inadequate insulin production. Diabetes that starts in pregnancy; known as gestational diabetes, increases a person's risk of later being diagnosed with Type 2 diabetes (Jadon, 2024; Eizirik, 2020).

Postprandial hyperglycemia refers to the rapid-fire rise in blood glucose situations succeeding the consumption of a meal. It is accounted a crucial factor in the sequence of diabetes and its associated complexities. Lifted postprandial glucose situations contribute to oxidative stress, endothelial dysfunction, and inflammation, which play a meaningful role in the progression of cardiovascular complexities (Yu, M. G., 2024). Carbohydrates are broken down with the help of certain enzymes, specifically α-amylase and α-glucosidase, which allow for digestion to occur after we eat food high in carbohydrates. The enzymes break down complex carbohydrates into usable molecules of glucose for absorption (Sinha, D., 2015). Enzyme inhibition has been broadly regarded as a viable strategy for treating a variety of metabolic diseases including Diabetes. Enzymes are critical components in the regulation of biochemical processes; when these enzymes function improperly, it leads to disease progression. Consequently, inhibiting specific enzymes represents a focused method of managing disease conditions. In the case of diabetes, carbohydrates are digested with the help of enzymes, including α-amylase, α-glucosidase, and due to their role in regulating blood glucose levels after meals, have been isolated as important therapeutic targets (Verma, G.,2024)

The natural chemical compounds from plants have recently attracted much attention because they have many possible therapeutic uses and very few side effects. Plant based chemicals contain many biologically active molecules such as flavonoids, phenolic acids, and tannins; however flavonoids and phenolic compounds are most well known for their ability to inhibit enzymatic activity through their ability to act as antioxidants by binding to the active sites of enzymes (Sun, W.,2023). For example, quercetin, which is a plant based chemical and as a naturally occurring flavonoid, can be found in many fruits, vegetables, and medicinal plants such as onions, apples, and berries. There is a great deal of evidence to support the use of quercetin for its antioxidant, anti-inflammatory, and anti-diabetic properties (Sultana, B., 2008). Similar to quercetin, gallic acid, which is

a naturally occurring phenolic compound, is also widely distributed in many different plants, fruits, and medicinal plants; in particular, there is an abundance of data showing the usefulness of gallic acid for its powerful antioxidant properties, whereby gallic acid can neutralize free radicals (submost common examples) and, thus, reduce oxidative stress on a cell; therefore, since there is strong evidence now indicating that oxidative stress is one of the leading causes of diabetic progression, the therapeutic potential for using gallic acid for diabetes has also been recognized (Lakhanpal, P., 2007). Acarbose has been identified as the care standard for treating diabetes (oral anti-diabetic drug), as it has an α-glucosidase inhibition action (i.e., inhibits α-glucosidase enzyme activity). It had been introduced to manage type 2 diabetes mellitus by targeting postprandial hyperglycemia, which was considered a major contributor to long-term diabetic complications (Martin, A. E., 1996).

The use of combinations of bioactive compounds had emerged as a promising strategy in pharmacological research. Dissimilar to therapies that only have one compound, combination therapy provides researchers with an opportunity to determine the impacts of interaction amongst different molecules on each other. Synergistic effect was described when two compounds together produced a greater inhibitory or therapeutic impact than the sum of their individual actions Chaachouay, N., 2025).

There is a growing number of people diagnosed with diabetes, while current medication for this condition is limited thus creates a need for new ways to treat diabetes through the use of natually occuring compounds in food as an evidence-based approach to diabetes. The objective of this study was to determine the ability of these compounds as enzyme inhibitors both individually and together; therefore giving us an extensive amount of data on their potential for therapeutic use.

Materials and Methods

2.1 Chemicals and Drugs Materials

The chemicals and drugs required for the present study were obtained from the laboratory of Kalinga University. Routine laboratory chemicals including quercetin, gallic acid, α-amylase solution, soluble starch, DNSA reagent, phosphate buffer (pH 7.0), methanol and distilled water were available in the institutional laboratory and were used for the experimental work. The standard inhibitor Acarbose was procured in the form of commercially available tablets from a local pharmacy and was suitably processed and dissolved to prepare the required stock solution. The specific reagents required for the α-glucosidase inhibition assay, namely α-

glucosidase enzyme and p-nitrophenyl-α-D-glucopyranoside (pNPG) substrate, were not available in the laboratory. Therefore, these were procured from Sisco Research Laboratories. The order was placed on 25th February 2026 and the materials were received on 10th March 2026. All chemicals used in the study were of analytical grade and were used without further purification.

2.2 Methodology

Preparation of Stock Solutions of Test Compounds and Standard Drug

Stock solutions of the selected phytochemicals (Quercetin and Gallic Acid) as well as the reference antidiabetic drug (Acarbose) were prepared at a concentration of 1000 µg/mL to serve as the primary working solutions for subsequent enzyme inhibition assays (Wairata, 2022). The stock solutions were prepared under aseptic methods and stored in the appropriate manner with good laboratory practice to promote stability and reliability of results in the future.

Preparation and Use of Phosphate Buffer (pH 7.0)

In the present study, a phosphate buffer solution of pH 7.0 was employed (Krajewska, 1999). The buffer was not freshly prepared during the course of the experiments; instead, it was obtained from the laboratory stock maintained by the department. The buffer solution was initially made by mixing sodium dihydrogen phosphate (NaH_2PO_4) and disodium hydrogen phosphate (Na_2HPO_4) together in distilled water according to standard laboratory methods.

Standardization of Working Solutions

Solutions of Quercetin, Gallic Acid and Acarbose were made from each of their 1000 µg/mL stock solutions using a serial dilution method. The solutions were diluted with phosphate buffer at pH 7.0. The experiment had a limited number of replicates (n = 1). Therefore, all working solutions were prepared immediately prior to each assay to provide the highest possible degree of accuracy and reproducibility (Kamtekar, 2014).

Formulation and handling of Enzyme Solutions

The enzyme solutions required for the inhibition assays were prepared according to the availability and nature of the enzymes employed in the study (Smirnovienė, 2021). The α-amylase enzyme solution was readily

available in the laboratory of the college and was directly utilized for the experimental work without any further preparation. α-glucosidase enzyme was not available in the laboratory and was therefore procured in powder form from Sisco Research Laboratories (SRL), India. A fresh α glucosidase (Maltase) ex. Yeast, 100U/mg enzyme powder was prepared immediately before the experiment to guarantee maximum activity.

Composition of Substrate Solutions

To produce clear starch solution, starch was dissolved in distilled water with controlled heat and constant stirring. Weighing starch powder, little by little it added to distilled water. Before the mixture was transparent, it was heated and stirred gently until a clear uniform solution was formed. The final clear and uniform starch solution was allowed to cool to room temperature, which gave it good stability for use later (Pucher, 1936). Because we did not have p-nitrophenyl-α-D-glucopyranoside (pNPG) in my College laboratory, I ordered pNPG from Sisco Research Laboratories (SRL), India, in a total quantity of 100 mg. pNPG is used for the α-glucosidase inhibition assay and I prepared it fresh (in phosphate buffer), so I could maintain its activity and reproducibility during testing. The necessary amount of pNPG was weighed out accurately and dissolved in phosphate buffer (Kito, M., 1996). When preparing this solution, I was careful to ensure that no particulate was formed during the dissolving process. I always prepared this solution immediately prior to use to ensure that it would not degrade.

α-Amylase Inhibition Assay

Measured aliquot of compound was introduced into the reaction system. Pipetted 200 µL of test sample into a clean test tube. Introduced α amylase to initiate potential inhibitory interaction. Added 200 µL of α amylase enzyme solution. Ensured optimal pH and ionic strength for enzymatic activity. Added 400 µL of phosphate buffer. Allowed enzyme and inhibitor interaction under physiological conditions. Incubated mixture at 37°C for 10 minutes. Provided starch as the enzymatic substrate. Added 200 µL of starch solution. Permitted enzymatic hydrolysis of starch in presence of inhibitors. Incubated mixture for 10 minutes at 37°C. Stopped enzymatic activity by chemical denaturation. Added 400 µL of DNSA reagent. Facilitated color development through reduction of DNSA. Heated mixture in boiling water bath for 5 minutes. Stabilized reaction mixture and quantified absorbance. Allowed mixture to cool. Measured absorbance at 540 nm using spectrophotometer (Nyambe-Silavwe, 2015).

α-Glucosidase Inhibition Assay

The assay began with the preparation of the test solution. Pipetted 200 µL of test sample into a clean test tube. The enzyme was introduced to initiate potential interaction. Added 200 µL α glucosidase enzyme solution. Phosphate buffer was used to maintain optimal pH. Added 400 µL phosphate buffer to the mixture. The mixture was allowed to equilibrate. Incubated at 37°C for 10 minutes. The enzymatic reaction was initiated with pNPG. Added 200 µL pNPG substrate. Reaction was allowed to proceed. Incubated at 37°C for 20 minutes. The extent of hydrolysis was quantified. Measured absorbance at 405 nm using a spectrophotometer (Iqbal, S., 2013).

Results

1 α-Amylase Inhibition Study

TABLE 1: Spectrophotometric Determination of Alpha-Amylase Inhibitory Activity

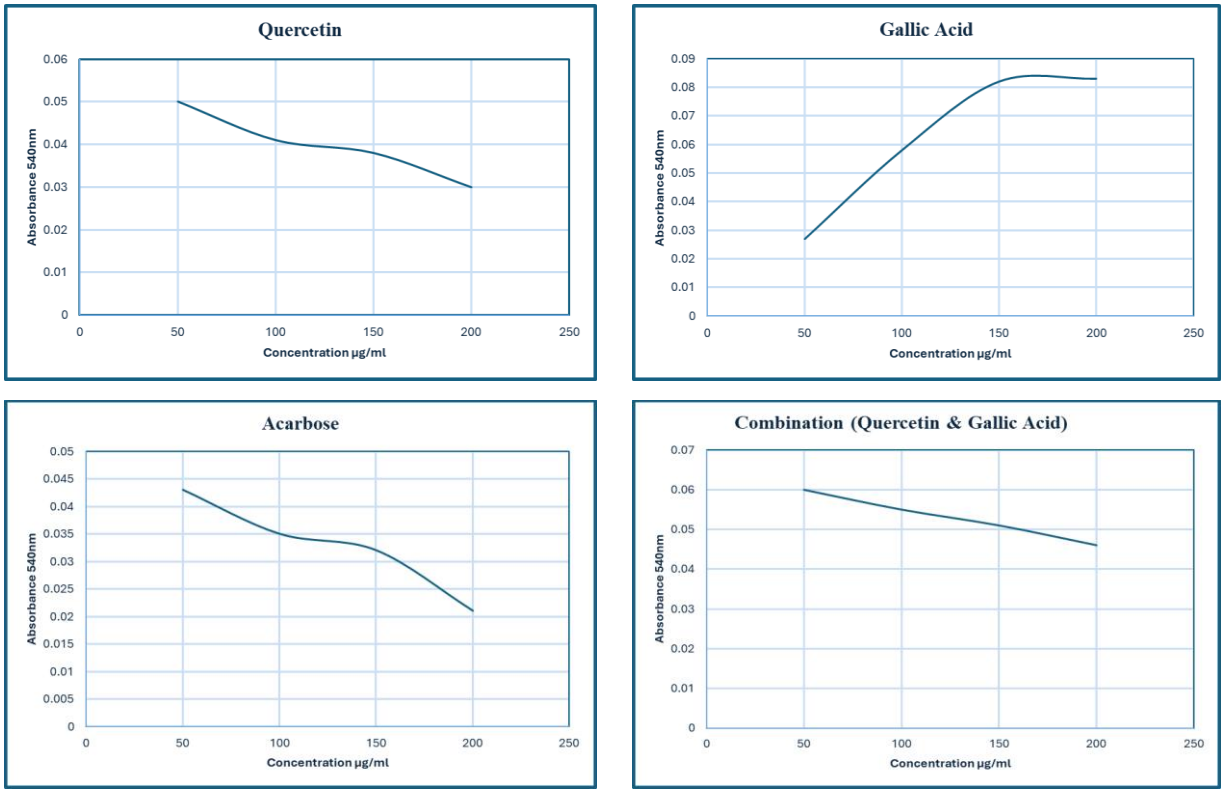
No.	Sample Name	Wavelength (nm)	Absorbance	Transmittance (%T)	Sample Energy	Reference Energy	Test Date	Test Time
1	Blank	540	0	100	21260	25903	3/20/2026	3:11:06 PM
2	50 Q	540	0.05	89.1	18975	25953	3/20/2026	3:13:49 PM
3	100 Q	540	0.041	90.9	20311	27228	3/20/2026	3:15:29 PM
4	150 Q	540	0.038	91.2	19866	27028	3/20/2026	3:16:55 PM
5	200 Q	540	0.03	93.3	20637	26958	3/20/2026	3:17:55 PM
6	50 G	540	0.027	94.1	20712	26839	3/20/2026	3:19:38 PM
7	100 G	540	0.058	87.5	19292	26888	3/20/2026	3:20:28 PM
8	150 G	540	0.082	82.9	18367	27019	3/20/2026	3:21:15 PM
9	200 G	540	0.083	82.6	18401	27140	3/20/2026	3:22:24 PM
10	50 A	540	0.043	90.7	20242	27214	3/20/2026	3:23:25 PM
11	100 A	540	0.035	92	21798	27216	3/20/2026	3:24:37 PM
12	150 A	540	0.032	93	20877	27374	3/20/2026	3:25:31 PM
13	200 A	540	0.021	95.3	21297	27230	3/20/2026	3:26:21 PM
14	50 C	540	0.06	87	19422	27204	3/20/2026	3:27:06 PM
15	100 C	540	0.055	88.2	19754	27312	3/20/2026	3:28:06 PM
16	150 C	540	0.051	89	19151	27487	3/20/2026	3:29:28 PM
17	200 C	540	0.046	90	19427	27540	3/20/2026	3:30:20 PM
18	Control	540	0.109	77.8	17554	27509	3/20/2026	3:31:19 PM

Table 2: % inhibition at different concentration in α-amylase assay

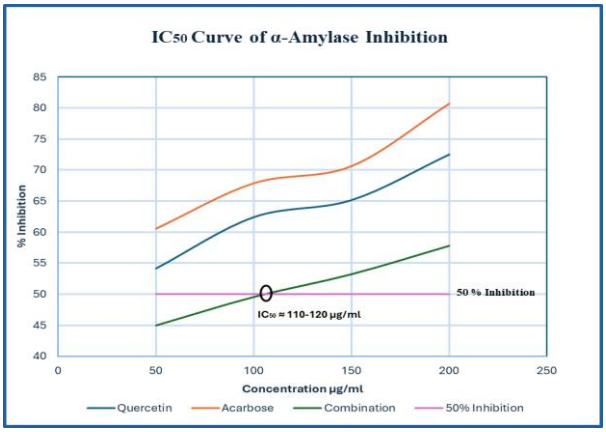
Sample	Concentration	Absorbance	Corrected	% Inhibition
Control	0	0.109	0.109	0
Blank	0	0	0	0
Quercetin	50	0.05	0.05	54.12%
Quercetin	100	0.041	0.041	62.38%
Quercetin	150	0.038	0.038	65.13%
Quercetin	200	0.03	0.03	72.47%

Gallic Acid	50	0.027	0.027	75.22%
Gallic Acid	100	0.058	0.058	46.78%
Gallic Acid	150	0.082	0.082	24.77%
Gallic Acid	200	0.083	0.083	23.85%
Acarbose	50	0.043	0.043	60.55%
Acarbose	100	0.035	0.035	67.88%
Acarbose	150	0.032	0.032	70.64%
Acarbose	200	0.021	0.021	80.73%
Combination	50	0.06	0.06	44.95%
Combination	100	0.055	0.055	49.54%
Combination	150	0.051	0.051	53.21%
Combination	200	0.046	0.046	57.79%

Graphical representation of absorbance at different concentrations



Graphical representation of IC_{50} curve of α-amylase inhibition



The IC_{50} values were found to be approximately 60–70 µg/mL for Acarbose, 90–100 µg/mL for Quercetin and 110–120 µg/mL for the combination.

2 α-Glucosidase Inhibition Study

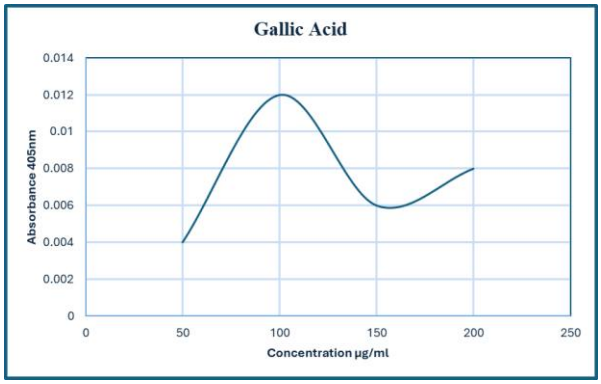
Table 3: Spectrophotometric Determination of α-Glucosidase Inhibitory Activity

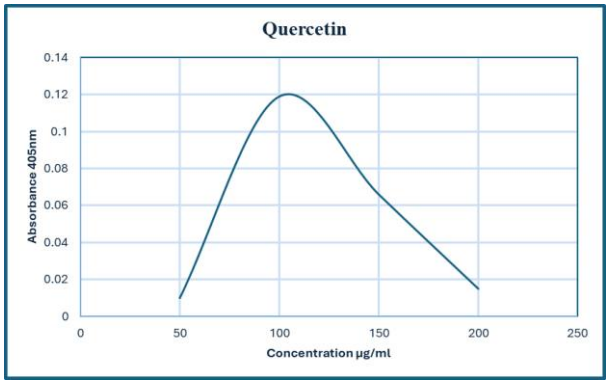
No.	Sample Name	Wavelength (nm)	Absorbance	Transmittance (%T)	Sample Energy	Reference Energy	Test Date	Test Time
1	Blank	405	0	100	17381	20165	3/23/2026	2:53:03 PM
2	50 Q	405	0.01	97.8	17428	20052	3/23/2026	3:01:22 PM
3	100 Q	405	0.119	76.1	13620	20020	3/23/2026	3:02:17 PM
4	150 Q	405	0.066	85.9	15320	20009	3/23/2026	3:03:05 PM
5	200 Q	405	0.015	96.7	17200	20012	3/23/2026	3:05:45 PM
6	50 G	405	0.004	99	17607	20008	3/23/2026	3:06:49 PM
7	100 G	405	0.012	97.3	17305	19995	3/23/2026	3:07:54 PM
8	150 G	405	0.006	98.5	17491	19970	3/23/2026	3:10:58 PM
9	200 G	405	0.008	98.2	17458	19992	3/23/2026	3:11:46 PM
10	50 A	405	0.001	99.8	17704	19960	3/23/2026	3:12:40 PM
11	100 A	405	0.002	99.5	17662	19963	3/23/2026	3:15:55 PM
12	150 A	405	0.002	99.5	17672	19990	3/23/2026	3:16:56 PM
13	200 A	405	0.001	99.8	17724	19990	3/23/2026	3:17:46 PM
14	50 C	405	0.013	97.1	17232	19958	3/23/2026	3:18:33 PM
15	100 C	405	0.061	86.9	15473	19964	3/23/2026	3:19:20 PM
16	150 C	405	0.056	88	15649	19966	3/23/2026	3:20:15 PM
17	200 C	405	0.075	84.2	15001	19970	3/23/2026	3:21:14 PM
18	Control	405	0.002	99.5	17666	19972	3/23/2026	3:22:14 PM

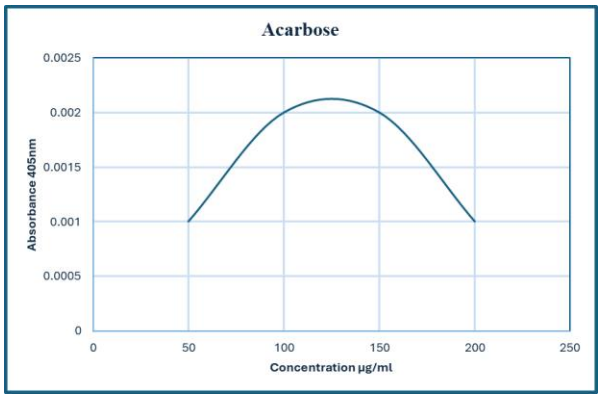
Table 4: % inhibition at different concentration in α-glucosidase assay

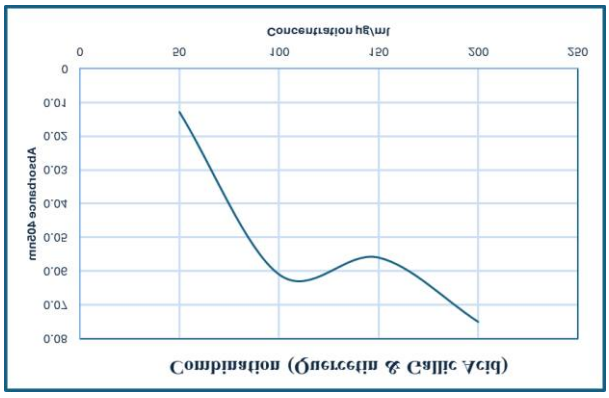
Sample	Concentration	Absorbance	Corrected	% Inhibition
Control	0	0.002	0.002	0
Blank	0	0	0	0
Quercetin	50	0.01	0.01	-400.00%
Quercetin	100	0.119	0.119	-5850.00%
Quercetin	150	0.066	0.066	-3200.00%
Quercetin	200	0.015	0.015	-650.00%
Gallic Acid	50	0.004	0.004	-100.00%
Gallic Acid	100	0.012	0.012	-500.00%
Gallic Acid	150	0.006	0.006	-200.00%
Gallic Acid	200	0.008	0.008	-300.00%
Acarbose	50	0.001	0.001	50.00%
Acarbose	100	0.002	0.002	0.00%
Acarbose	150	0.002	0.002	0.00%
Acarbose	200	0.001	0.001	50.00%
Combination	50	0.013	0.013	-550.00%
Combination	100	0.061	0.061	-2950.00%
Combination	150	0.056	0.056	-2700.00%
Combination	200	0.075	0.075	-3650.00%

Graphical representation of absorbance at different concentrations









Discussion

In the current study, we have determined how much of an effect quercetin, gallic acid, both of these items together in combination with each other and the control standard drug acarbose did in inhibiting α-amylase enzyme as well as α-glucosidase enzyme via spectrophotometric methods. For both assays conducted those data showed distinct patterns of inhibition against both enzymes tested. The inhibition assay conducted on the alpha amylase enzyme followed a trend and was consistent with all data gathered (Zinjarde, S. S., 2011). The control showed an absorbance reading of 0.109 therefore indicating the presence of maximum activity of the enzyme without the presence of any inhibitor(s). However, once tested samples also had absorbance readings that were lower when compared to the maximum (control). The decrease was basically due from the inhibition produced by each compound tested against the above-mentioned enzyme (α-amylase). At the same time with regard to quercetin and acarbose, a concentration-dependent increase was observed with percentage (%) inhibition indicating that as the concentration of each increased so too did their effectiveness of inhibiting the enzyme. There was a high level of inhibition with the combination demonstrating moderate levels of inhibition however less than both acarbose & quercetin when separately compared to either acarbose or quercetin alone respectively (Zhang, B. W., 2017). Data for gallic acid did not yield a clear concentration-dependent increase in percentage (%) inhibition however showed inconsistent and varying values due to lack of consistent increases in inhibition as concentration was increased and thereby showing decreases in other concentration ranges indicating no discernible dose-response effect was seen with gallic acid during the current analysis conducted. Due to the lack of or irregularity in pattern of data obtained, estimation of an IC_{50} value was not possible. On the contrary, the absence of a solid and trustworthy inhibitory trend was observed in α-glucosidase inhibition activity. Unlike the α-amylase assay where the control absorbance was greater than most test sample values, the control absorbance for this α-glucosidase assay was lower than that of many test samples. The control is usually the highest in absorbance measurement (maximum enzyme activity). However, in this experiment, some of the test compound and the standard drug absorbance was similar to or greater than the control. The inversion of these expected trends suggests that the data collected in this α-glucosidase show possibly inaccurate results (Topuz, Ş., 2025).

Several factors could account for the low reliability and consistency associated with α-glucosidase. For example, enzyme activity could have been variable from assay to assay or improper incubation conditions; pipetting error during the addition of reagents; or enzyme or substrate instability. Additionally, since the assay was run on n=1 (one reading), the lack of replicate measurements would contribute to additional variability and consequently low reliability of the data.

The enzymes may differ in their specificities for different substrates. The tested materials had a strong affinity for α-amylase, however they may not have very much affinity for α-glucosidase under the present experimental conditions. It is possible that the abnormal test/control data is primarily due to experimental limitations rather than true biological inactivity, because this information provides evidence of limit and significant deviation from expected response (Kaur, R., 2014).

The results of the study support that both quercetin and acarbose are potent inhibitors of α-amylase in a concentration dependent manner but indicate that there are inconsistencies in the results of the α-glucosidase assays and therefore cannot be accurately interpreted. In order to further evaluate the inhibitory potential of the tested materials on α-glucosidase, it will be necessary to conduct further studies with appropriate replication and more optimized experimental conditions.

Conclusion

The study showed that Acarbose had the strongest α-Amylase Inhibitory (IA) action, with Quercetin having a considerable amount of α-Amylase IAA too. When both compounds were combined, α-amylase IA exhibited moderate levels of inhibition; however, gallic acid did not show any consistency in its ability to inhibit α-amylase. The α-glycosidase assay results were imprecise due to experimental setup variables. There was inhibition of α-amylases in the data; however, further work is needed to substantiate the results from this experiment.

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Conflict of Interest

The authors declare that there are no conflict of interest related to this work.

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